

Workshop Report

Collaborators

Name	Institution
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Introduction

Describe the topic of your report here. What question are you trying to answer? Why is it relevant?

Data

Briefly describe the dataset you are using: where it comes from, how many observations it has, and what variables are included. All data pre-processing must be described in words and in the code

```
data(wcgs, package="faraway")
wcgs = na.omit(wcgs) # Comment on how many participants were removed!
descTab = wcgs %>% group_by(chd) %>%
  summarise(mAge = mean(age),
            sdAge = sd(age),
            mChol = mean(chol, na.rm = T),
            sdChol = sd(chol, na.rm = T))
xtable(descTab, digits = 0)
```

```
% latex table generated in R 4.5.1 by xtable 1.8-4 package
% Tue May 26 16:01:42 2026
\begin{table}[ht]
\centering
\begin{tabular}{rlrrrr}
\hline
& chd & mAge & sdAge & mChol & sdChol \\
\hline
1 & no & 46 & 5 & 224 & 42 \\
2 & yes & 49 & 6 & 250 & 50 \\
\hline
\end{tabular}
\end{table}
```

	CHD	<i>M</i> Age	<i>SD</i> Age	<i>M</i> Chol	sdChol
1	no	46.08	5.46	224.26	42.22
2	yes	48.49	5.80	250.07	49.40

Add a short description of the variables that are most relevant to your analysis.

Colour palette: <https://htmlcolorcodes.com/>

```
ggplot(data = wgs)+
  geom_histogram(aes(x = chol), col = "black", fill = "lightblue")+
  theme_bw() +
  theme(panel.background = element_rect(fill = "#FFDCBF"),
        plot.background = element_rect(fill = "#FFDCBF"))
ggplot(data = wgs)+
  geom_histogram(aes(x = sdps), col = "black", fill = "lightblue")+
  theme_bw() +
  theme(panel.background = element_rect(fill = "#FFDCBF"),
        plot.background = element_rect(fill = "#FFDCBF"))
```

The sample distributions for the independent variables cholesterol and systolic blood pressure (BP) are displayed in Figure 1.

Methods/Analysis/Statistical theory

Explain the method or approach you are using to analyse the data.

$$f(Y_t|S_{1:(t-1)}, Y_{1:(t-1)}) = f(Y_t|S_t) \quad (1)$$

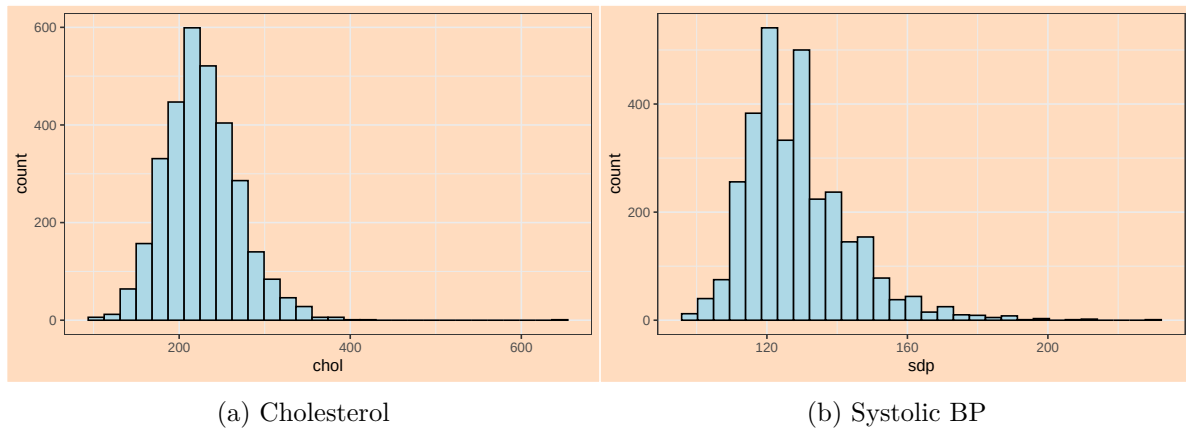


Figure 1: Histograms

The random variable Y only depends on the hidden state S , as indicated in Equation 1.

Sometimes Latex does not render nicely within the dollar-dollar environment. Then you might use raw Latex code instead.

“Manual” latex: <https://editor.codecogs.com/>

Only number equations that are referred to in the text.

$$\begin{aligned}
 \text{logit}(p) &= \log(\text{odds}) \\
 &= \log\left(\frac{p}{1-p}\right) \\
 &= \eta \\
 &= \beta_0 + \beta_1 x_1
 \end{aligned}$$

Sometimes it can be nice to include pseudocode to explain a function. Pseudocode is code that is not functional but describes the general flow

```

MLEfunction {
  initialise betaVector
  calculate logLikelihood

  maximising loop {
    update betaVector
    calculate logLikelihoodNew
  }
  return(betaVector)
}

```

Results

Summarise what you found. You can reference numbers, tables, or plots produced above.

```
mod1 <- glm(chd ~ height , family = binomial, wgs)
summary(mod1)
```

Call:

```
glm(formula = chd ~ height, family = binomial, data = wgs)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.08506	1.82164	-2.243	0.0249 *
height	0.02375	0.02604	0.912	0.3617

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1769.2 on 3139 degrees of freedom
Residual deviance: 1768.3 on 3138 degrees of freedom
AIC: 1772.3

Number of Fisher Scoring iterations: 5

```
mod2 <- glm(chd ~ height + cigs, family = binomial, wgs)
summary(mod2)
```

Call:

```
glm(formula = chd ~ height + cigs, family = binomial, data = wgs)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.219165	1.852850	-2.277	0.0228 *
height	0.021019	0.026495	0.793	0.4276
cigs	0.023584	0.004059	5.810	6.24e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1769.2 on 3139 degrees of freedom
Residual deviance: 1736.3 on 3137 degrees of freedom
AIC: 1742.3

Number of Fisher Scoring iterations: 5

```
mod3 <- glm(chd ~ height + cigs + chol, family = binomial, wgs)  
summary(mod3)
```

Call:

```
glm(formula = chd ~ height + cigs + chol, family = binomial,  
     data = wgs)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.345960	1.959430	-4.259	2.05e-05 ***
height	0.040222	0.027095	1.484	0.138
cigs	0.020865	0.004174	4.999	5.77e-07 ***
chol	0.011950	0.001455	8.216	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1769.2 on 3139 degrees of freedom
Residual deviance: 1667.1 on 3136 degrees of freedom
AIC: 1675.1

Number of Fisher Scoring iterations: 5

```
deviances = anova(mod1, mod2, mod3)  
deviances
```

Analysis of Deviance Table

Model 1: chd ~ height

Model 2: chd ~ height + cigs

Model 3: chd ~ height + cigs + chol

```

      Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1         3138      1768.3
2         3137      1736.3  1    32.011 1.533e-08 ***
3         3136      1667.1  1    69.241 < 2.2e-16 ***
---

```

```

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# xtable(deviances)

```

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	3138	1768.34			
2	3137	1736.33	1	32.01	0.0000
3	3136	1667.08	1	69.24	0.0000

```

xtable(mod2)

```

```

% latex table generated in R 4.5.1 by xtable 1.8-4 package
% Tue May 26 16:01:43 2026
\begin{table}[ht]
\centering
\begin{tabular}{rrrrrr}
\hline
& Estimate & Std. Error & z value & Pr(>|z|) & \\
\hline
(Intercept) & -4.2192 & 1.8528 & -2.28 & 0.0228 & \\
height & 0.0210 & 0.0265 & 0.79 & 0.4276 & \\
cigs & 0.0236 & 0.0041 & 5.81 & 0.0000 & \\
\hline
\end{tabular}
\end{table}

```

The results of the best model is presented in Table 2

It is always fine to run 10 models without reporting the exact coefficients for all of them! But make sure to report the relevant statistics!

Table 2: Logistic regression model predicting coronary heart disease

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.5016	1.8419	-2.44	0.0145
height	0.0252	0.0263	0.96	0.3383
cigs	0.0231	0.0040	5.72	0.0000

Discussion

In the results you only report your findings, the discussion is for the interpretation and what the results mean. For instance, here you can discuss the relative importance of covariates, or why the findings are (not) surprising. Be sure to include some comments on limitations and future research ideas if you have any

Conclusions

Summarise the main findings of your analysis

References

List any sources, datasets, or packages you used.